Task 6 — Create a Strong Password and Evaluate Its Strength

Objective: To understand what makes a password strong and test it using password strength checkers.

Tools Used:

• passwordmeter.com
• howsecureismypassword.net

Step 1: Create Multiple Passwords

Password	Complexity Type	Composition Used
password123	Weak	Lowercase + numbers
Pass@2025	Medium	Uppercase + lowercase + num + symbol
G!tH#b_Intern2025	Strong	Mixed case + numbers + multiple symbols
T!mE$Is@Precious2025!!	Very Strong	Long length + mixed case + symbols + numbers

Step 2: Test on Strength Checkers

Password	Score (%)	Crack Time (Estimate)	Feedback
password123	20%	< 1 second (brute force)	Too common, predictable
Pass@2025	60%	Few hours	Better, but short length
G!tH#b_Intern2025	85%	Hundreds of years	Strong, good complexity
T!mE$Is@Precious2025!!	100%	Millions of years	Excellent, very strong

Step 3: Best Practices Learned

1. Length matters: Longer passwords are much harder to brute-force.
2. Complexity is key: Use uppercase, lowercase, numbers, and special characters.
3. Avoid common words: Dictionary-based passwords are easy to crack.
4. Symbols & randomness: Adding unusual symbols improves security.
5. Passphrases: Long passphrases (e.g., Sunshine!River^2025!BlueSky) are both strong and easy to remember.

Step 4: Research on Password Attacks

• Brute Force Attack: Attacker tries all possible combinations.
• Dictionary Attack: Uses common words/passwords.
• Credential Stuffing: Reuses leaked credentials.
• Phishing: Tricks user into revealing password.

Step 5: Summary

Password complexity and length directly impact resistance against brute-force and dictionary attacks.
Strong passwords should be:
- At least 12–16 characters long
- Contain upper/lowercase letters, numbers, and symbols
- Not reused across multiple sites
- Stored in a password manager
- Combined with Multi-Factor Authentication (MFA)

Interview Prep Questions & Answers

1. **What makes a password strong?** A strong password is long, complex, unpredictable, and unique.

2. **What are common password attacks?** Brute force, dictionary attack, credential stuffing, phishing.

3. **Why is password length important?** It exponentially increases possible combinations.

4. **What is a dictionary attack?** An attack using precompiled word lists.

5. **What is MFA?** Adding a second authentication factor beyond the password.

6. **How do password managers help?** They generate, store, and autofill strong unique passwords.

7. **What are passphrases?** Long sequences of words that are easier to remember and secure.

8. **What are common mistakes?** Using personal info, reusing passwords, short/dictionary words.

Outcome: I now understand how to create strong passwords, test their strength, and defend against common password attacks.